

1. Question 1

The minimum number of 2 to 1 multiplexers required to realize a 4 to 1 multiplexer is

Pick **ONE** option

☐ 2

☐ 1

☐ 4

☐ 3

Solve question 37

Clear Selection

3. Question 3

What is the output of following program in javascript
console.log(typeof(String)
console.log(typeof(object)
console.log(typeof(Array)
console.log(typeof(function a(){}))

Pick **ONE** option

☐ object
object
object
function

☐ function
undefined
function
function

☒ function
object
object
function

☐ string
object
array
function

Clear Selection

4. Question 4

Which DNS security would you choose to use, inside highly secure corporate network behind strict firewall rules, that hides query information during transmission?

Pick **ONE** option

☐ DNSSEC

☐ DoT

☒ DoH

☐ Any of Above

Clear Selection

5. Question 5

Consider the set of integers from 1 to 10, inclusive. Let x be a randomly selected integer from the set and y be another randomly selected integer from the same set. If $x+2y>15$, what is the probability that the product of x and y is greater than 40?

Pick **ONE** option

☐ 0.25

☐ 0.66

☒ 0.33

☐ 0.5

Clear Selection

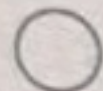
6. Question 6

Which of the following is not correct approach to develop algorithm?

Pick **ONE** option



Brute Force



Dynamic Programming



Randomised Greedy



Back Tracking

Clear Selection



7. Question 7

A university wants to enable Apple TV streaming for all their classrooms which uses mDNS protocol. What type of traffic and class of IP address will this belong to?

Pick **ONE** option

☐ Its Class D, used for broadcast traffic

☐ Its Class B, used for Local Area Network

☒ Its Class D, used for multicast traffic

☐ Its Class C, used for Public IPs

Clear Selection

8. Question 8

What is the output of the following code?

```
numbers = [1, 2, 3, 4, 5]
```

```
print([x for x in numbers if any(x % y == 0 for y in range(2, x))])
```

Pick **ONE** option

☐ [4]

☐ [2,3,5]

☐ [1,2,3,5]

☒ [1,3,5]

Clear Selection

9. Question 9

Adding a new input variable to a pre-existing regression model is guaranteed to:

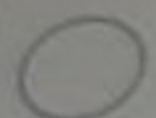
Pick **ONE** option



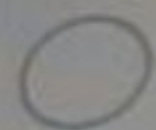
☒ Increase R^2



Increase Adj- R^2



Increase p -value of the F -statistic in the ANOVA model



Decrease p -value of the F -statistic in the ANOVA model

9. Question 9

Criteria for OSPF DR and BDR election?[Choose two]

Pick **ONE OR MORE** options

☐

Router Priority

☒

Router ID

☐

Loopback Address

☐

OSPF Area ID

Clear Selection

13. Question 13

You are building multiboot system with various Windows and Linux distros. Which bootloader would you use, keeping support for both UEFI and Legacy boot?

Pick **ONE** option

☐ systemd-boot

☒ GNU GRUB

☐ BSD Loader

☐ Any of above

Clear Selection

19. Question 19

"Program runs in 20 seconds on a computer Draco, which has 400MHz clock. What should be the clock rate of computer hercules to run the same program in 10 seconds. Given Hercules takes 2 times as many clock cycles as draco for this program. "

Pick **ONE** option

☐ 1600Mhz

☐ 800Mhz

☐ 2000Mhz

☐ 400Mhz

Clear Selection

20. Question 20

The minimum number of 2 to 1 multiplexers required to realize a 4 to 1 multiplexer is

Pick **ONE** option

☐ 2

☐ 1

☐ 4

☐ 3

Clear Selection

HackerRank Confidential

21. Question 21

What will happen if the switch interface is configured with Full Duplex and 1 Gbps rate, while the server side is configured with half duplex and 1 Gbps rate, and there is inbound and outbound traffic at 1 Gbps?

Pick **ONE OR MORE** options

☒ Packets drop

☐ No impact

☐ High rx/tx load

☐ CRC

Clear Selection

21. Question 21

What will happen if the switch interface is configured with Full Duplex and 1 Gbps rate, while the server side is configured with half duplex and 1 Gbps rate, and there is inbound and outbound traffic at 1 Gbps?

Pick **ONE OR MORE** options

☒

Packets drop

☐

No impact

☐

High rx/tx load

☐

CRC

Clear Selection

22. Question 22

Which option below does not reduce network congestion?

Pick **ONE** option

☐ Adjusting TCP/IP settings to balance packet send/request speeds

☒ Relying on UDP to reduce packets transmission

☐ Employ multi-hop routing with path redundancy

☐ Use a CDN (Content Delivery Network)

Clear Selection

22. Question 22

Which option below does not reduce network congestion?

Pick **ONE** option

☐ Adjusting TCP/IP settings to balance packet send/request speeds

☒ Relying on UDP to reduce packets transmission

☐ Employ multi-hop routing with path redundancy

☐ Use a CDN (Content Delivery Network)

Clear Selection

23. Question 23

Eight bit Ripple up counter with a modules of 256 is holding the count 01111111. What will be the count after 135 clks

Pick **ONE** option

☒ 0000110

☐ 0000101

☐ 00011000

☐ 00110010

Clear Selection

30. Question 30

A developer is working on a new application that is supposed to process tasks. The application has three modules: Module A, Module B, and Module C. Each module processes tasks at different speeds and has different constraints:

- Module A can process 60 tasks per hour but cannot operate for more than 6 continuous hours without an hour of maintenance.
- Module B can process 40 tasks per hour and can operate non-stop.
- Module C can process 25 tasks per hour but can only operate for a total of 8 hours in a 24-hour period.

The developer wants to maximize the number of tasks processed in a 24-hour period. Assume there is an unlimited number of tasks to be processed and no time is lost for switching between modules.

How many tasks can the developer process in a 24-hour period by optimally using the modules?

Pick **ONE** option

- ☒ 12500
- ☐ 2394
- ☐ 3244
- ☐ 1M

Clear Selection

31. Question 31

Which of below is not supported by TCP?

Pick **ONE** option

☐ Data transmission in sequence

☒ Broadcast of data

☐ Data Re-transmission

☐ Congestion control

Clear Selection

32. Question 32

Assume that the height of all eighth grade student in a school follows a Normal distribution. The teacher randomly selects 20 students, and from this sample, builds a confidence interval for the estimate of the average height of all eighth grade students. She finds it to be (155.24, 164.77). This means that:

Pick **ONE** option

- ☐ 95% of the students will have heights within this range
- ☐ We are 95% certain that a randomly selected student will have a height within this range
- ☒ We are 95% certain that the true average height of students lies within this range
- ☐ none of the above.

Clear Selection

33. Question 33

A statistician is asked to help a court determine the chance of innocence of an accused. Witnesses in this theft case provide the following description.

They saw a Magenta coloured van driven by a man in his early 30s with grey-beard, force breaking into a store using a rod. The criminal in question used his left hand to break open the store. The accused had his face covered with an opaque mask while attempting to commit burglary.

The statistician quotes the following events and associated probabilities.

Assume these numbers to be factually correct and that the van belonged to the criminal (not necessarily the accused).

Owning a Magenta Van -- 1 in 6000

Man to be in his early 30s -- 1 in 5

Man with grey beard -- 1 in 10

Man being left-handed -- 1 in 10

After quoting these stats, the statistician then goes on to state the probability of innocence of the accused based on the above facts. Pick the correct value of probability

Pick **ONE** option

☐ $(1/6000) * (1/5) * (1/10) * (1/10)$

☒ $1 - [(1/6000) * (1/5) * (1/10) * (1/10)]$

☐ $1 - [(1/6000) + (1/5) + (1/10) + (1/10)]$

☐ Insufficient Information

Clear Selection

34. Question 34

"The minimum number of flip-flops required to construct a mod-50 counter? "

Pick **ONE** option

☐ 5

☒ 6

☐ 7

☐ 8

Clear Selection

35. Question 35

What are the three parameters a router considers for choosing a route?

Pick **ONE** option

☒ Administrative distance, metrics, longest prefix match

☐ Link bandwidth, Administrative distance, shortest prefix match

☐ Link bandwidth, Link delay, next-hop distance

☐ Routing protocol used, number of hops, link bandwidth

Clear Selection